Graphic Era HILL UNIVERSITY

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B. Tech ECE 5th Semester
Name of the Paper: Digital System Design using VHDL
Paper Code: TEC 502
Time: 3-hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Explain CO1
(i) EDA Tool
(ii) 8 to 3 Encoder entity declaration using VHDL

b. Explain the working of demultiplexer. Write the VHDL code of 1 to 4 demultiplexer. CO1

c. Explain CO1
(i) VHDL Design Flow
(ii) Architecture of Full adder using Data flow model

Q2. (2X10=20 Marks)

a. Explain CO2
(i) Pre-Defined Data Attributes
(ii) Different arrays used in VHDL

b. Explain CO2
(i) Std_ULogic Data Types values
(ii) Arithmetic operators

c. Write VHDL code to add two 8-bit unsigned numbers and store the result in a 9-bit signal. Explain shift operators. CO2

Q3. (2X10=20 Marks)

a. Explain and design D Flip Flop using IF and ELSEIF statement. CO3

b. Design and Explain 8:3 Encoder using WHEN/ELSE statement. CO3

c. Explain Signals, variables and Constant and write their syntax also. CO3

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Q4. (2X10=20 Marks)

a. Draw the state graph of SR and T flip flop. CO4

b. Explain different components of SM chart. Draw and explain the SM Chart for Binary Multiplier. CO4

c. Explain the working of serial adder with accumulator. CO4

Q5. (2X10=20 Marks)

a. Explain the working of Transmitter and Receiver in UART block diagram. CO5

b. Explain the block diagram of MC68HC05 microcontroller. CO5

c. Explain floating-point multiplication with the help of example. CO5